

Project Design Phase-II

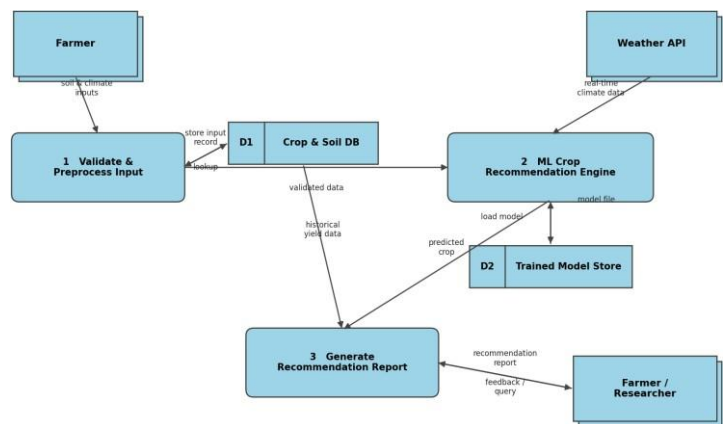
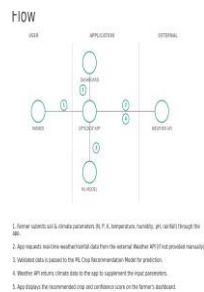
Data Flow Diagram & User Stories

Date	27 June 2026
Project Name	OptiCrop:Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(OptiCrop Process Flow\)](#)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Farmer (Mobile / Web User)	Registration	USN-1	As a farmer, I can register for the application using my mobile number, email, and password.	I can access my dashboard after registering.	High	Sprint-1
	Registration	USN-2	As a farmer, I will receive a confirmation OTP/email once I have registered for the application.	I can verify my account via OTP or confirmation email link.	High	Sprint-1
	Login	USN-3	As a farmer, I can log into the application using my registered email/mobile number and password.	I can access my dashboard after successful login.	High	Sprint-1
	Soil & Environmental Data Input	USN-4	As a farmer, I can enter my soil values (N, P, K, pH) and climate values (temperature, humidity, rainfall) manually.	The input form validates and accepts all required fields.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
	Soil & Environmental Data Input	USN-5	As a farmer, I can let the app auto-fetch local weather data instead of entering it manually.	Weather data is fetched and pre-filled correctly for my location.	Medium	Sprint-2
	Crop Recommendation Generation	USN-6	As a farmer, I can submit my data and receive the top recommended crop along with a confidence score.	Recommendation is displayed within a few seconds of submission.	High	Sprint-2
	Crop Recommendation Generation	USN-7	As a farmer, I can check the suitability score for a specific crop that I choose.	Suitability score and a short explanation are displayed.	Medium	Sprint-3
	Notifications & Alerts	USN-8	As a farmer, I receive an email/SMS with my crop recommendation report.	The report is delivered to my registered email or mobile number.	Low	Sprint-3
Agricultural Researcher / Policymaker	Research & Analytics Dashboard	USN-9	As a researcher, I can view aggregated crop-environment trends across regions and seasons.	Dashboard displays charts that can be filtered by region, season, or crop type.	Medium	Sprint-3
	Research & Analytics Dashboard	USN-10	As a researcher, I can export crop-environment analysis reports in PDF/CSV format.	Report downloads correctly with the selected filters applied.	Low	Sprint-4
Administrator	User Management	USN-11	As an administrator, I can view and manage registered farmer and researcher accounts.	Admin can view, edit, or deactivate user accounts.	Medium	Sprint-4
	Model Management	USN-12	As an administrator, I can upload or update the trained ML model used for crop recommendations.	New model file is validated and deployed without downtime.	Low	Sprint-4